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Subject: Programming with Java

ASSIGNMENT No: 2

TITLE : Constructors in Java

Problem Statement

Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Learning Objectives

To understand and implement default, parameterized, and copy constructors for object initialization.

Theory

Constructors are special member functions that are automatically invoked when an object is created. Java supports default constructors provided by the compiler, parameterized constructors for initializing objects with user-defined values and copy constructors (implemented manually) for creating duplicate objects. Constructors improve code readability and simplify object initialization while supporting object-oriented programming principles.

Programme

Exercise 1. Create a student class using default and parameterized constructors to initialize student name and roll number.

Answer:

```
Restaurant.java Student.java
J Student.java > Java > Student
1 class Student {
2
3     String name;
4     int rollNo;
5
6
7     Student() {
8         name = "Apurva";
9         rollNo = 1;
10    }
11
12    Student(String n, int r) {
13        name = n;
14        rollNo = r;
15    }
16
17    void display() {
18        System.out.println("Student Name: " + name);
19        System.out.println("Roll Number: " + rollNo);
20        System.out.println();
21    }
22
23    Run | Debug | Run main | Debug main
24    public static void main(String[] args) {
25
26        // Object using default constructor
27        Student s1 = new Student();
28
29        // Object using parameterized constructor
30        Student s2 = new Student("Rahul", 2);
31
32        s1.display();
33        s2.display();
34    }
```

Output

```
Problems 11 Output Debug Console Terminal Ports Spell Checker 2
-pack-jdk\java\latest\bin\java.exe --enable-preview -XX:+ShowCodeDetailsInExceptionMessages -cp "C:\Users\avnis\AppData\Roaming\Code
er\workspaceStorage\f69d52f9fd63939ec3885d040aad1893\redhat.java\jdt_ws\java folder new_c8a10cf9\bin" Student "
Student Name: Apurva
Roll Number: 1

Student Name: Rahul
Roll Number: 2
```

2. Develop a Mobile Phone Inventory System using different constructors to initialize mobile details and create duplicate object records.

Answer:

```
Restaurant.java Student.java Mobile.java
J Mobile.java > Language Support for Java(TM) by Red Hat > Mobile
1 class Mobile {
2     String brand;
3     String model;
4     double price;
5
6     // Default Constructor
7     Mobile() {
8         brand = "Samsung";
9         model = "Galaxy A15";
10        price = 18000;
11    }
12
13    // Parameterized Constructor
14    Mobile(String brand, String model, double price) {
15        this.brand = brand;
16        this.model = model;
17        this.price = price;
18    }
19
20    // Copy Constructor
21    Mobile(Mobile m) {
22        this.brand = m.brand;
23        this.model = m.model;
24        this.price = m.price;
25    }
26
27    // Method to display mobile details
28    void display() {
29        System.out.println("Brand      : " + brand);
30        System.out.println("Model      : " + model);
31        System.out.println("Price     : Rs. " + price);
32    }
33
34    Run | Debug | Run main | Debug main
35    public static void main(String[] args) {
36
37        // Object using Default Constructor
38        Mobile m1 = new Mobile();
39
40        // Object using Parameterized Constructor
41        Mobile m2 = new Mobile(brand: "Apple", model: "iPhone 15", price: 65000);
42
43        // Duplicate object using Copy Constructor
44        Mobile m3 = new Mobile(m2);
45
46        System.out.println(x: "Mobile 1 - Default Constructor");
47        m1.display();
48
49        System.out.println(x: "\nMobile 2 - Parameterized Constructor");
50        m2.display();
51
52        System.out.println(x: "\nMobile 3 - Duplicate Record");
53        m3.display();
54    }
```

Output:

```
Problems (11) Output Debug Console Terminal Ports Spell Checker (2)
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\avnis\OneDrive\Desktop\java folder new> cmd /C "C:\Users\avnis\AppData\Roaming\Code\User\globalStorage\pleiades.java-extension-pack-jdk\java\latest\bin\java.exe -cp "C:\Users\avnis\AppData\Roaming\Code\User\workspaceStorage\1f69d52f9d63939ec3885d848aad1893\redhat.java\jdt_ws\java folder new\bin" Mobile.java"
Mobile 1 - Default Constructor
Brand      : Samsung
Model      : Galaxy A15
Price     : Rs. 18000.0

Mobile 2 - Parameterized Constructor
Brand      : Apple
Model      : iPhone 15
Price     : Rs. 65000.0

Mobile 3 - Duplicate Record
Brand      : Apple
Model      : iPhone 15
Price     : Rs. 65000.0

C:\Users\avnis\OneDrive\Desktop\java folder new>
```

Conclusion:

Constructors are an important part of Java object-oriented programming because they help initialize objects when they are created. The default constructor provides default values, while the parameterized constructor allows the programmer to initialize an object with specific values. A copy constructor can be implemented manually to create a new object with the same values as an existing object. Using constructors makes programs easier to understand, reduces repeated initialization code, and improves code reusability. Thus, constructors provide a simple and efficient way to initialize and manage objects in Java.

LEET CODE:

